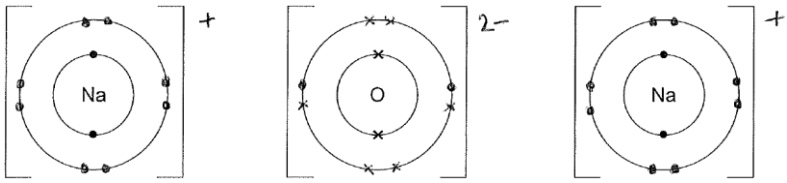


# HIGHER TIER ONLY QUESTIONS

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks available |     |     |       |       |      |
|----------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4        | (a) | (i)  |  |  <p>award (1) for 8 dots in outer shell of <b>both</b> sodium ions</p> <p>award (1) for 6 crosses and 2 dots in outer shell of oxide ion</p> <p>do not accept if extra electrons drawn in inner shell</p> <p>award (1) for + charge on sodium ions <b>and</b> 2- charge on oxide ion</p> <p>accept if + charge shown on only one sodium ion</p> |                 | 3   |     | 3     |       |      |
|          |     | (ii) |  | $4\text{Na} + \text{O}_2 \longrightarrow \boxed{2} \text{Na}_2\text{O}$ <p>award (1) for product</p> <p>award (1) for balancing <b>only if product correct</b></p>                                                                                                                                                                                                                                                                |                 | 2   |     | 2     | 1     |      |